

Question number	Answer	Additional guidance	Mark
3(a)(i)	Choose an item • C is the correct answer A is not the correct answer as the action potential is not caused by the movement of K^+ ions into the axon. B is not the correct answer as the action potential is not caused by the movement of K^+ ions out of the axon. D is not the correct answer as the action potential is not caused by the movement of Na^+ ions out of the axon.		(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	Choose an item. • A is the correct answer B is not the correct answer C is not the correct answer D is not the correct answer		(1)

Question number	Answer	Additional guidance	Mark
3(b)	A description that includes three of the following points: • receptor cells synapse with the sensory neurone (1) • (sensory neurone) synapses with relay neurone in grey matter of spinal cord (1) • (relay neurone) synapses with motor neurone (1) • motor neurone synapses with effector for response (1)	accept correctly labelled diagram	(3)

Question number	Answer	Additional guidance	Mark
3(c)	An explanation that includes four of the following points: • voltage gated (calcium) ion channels open (in pre synaptic membrane) causing calcium ions to (diffuse in / enter) (1) • causing vesicles (of neurotransmitter) to fuse with (pre synaptic) membrane (1) • {exocytosis/ release} of neurotransmitter (into synaptic cleft) (1) • which diffuses across and binds to receptors (on post synaptic membrane) (1) • causing {sodium channels to open / allowing sodium ions to enter} resulting in {depolarisation / impulse/action potential} (in post-synaptic neurone) (1)	ignore if the calcium ions are going into the wrong structure accept binds as eq fuse accept release of {neurotransmitter / correctly named neurotransmitter } into synapse	(4)

